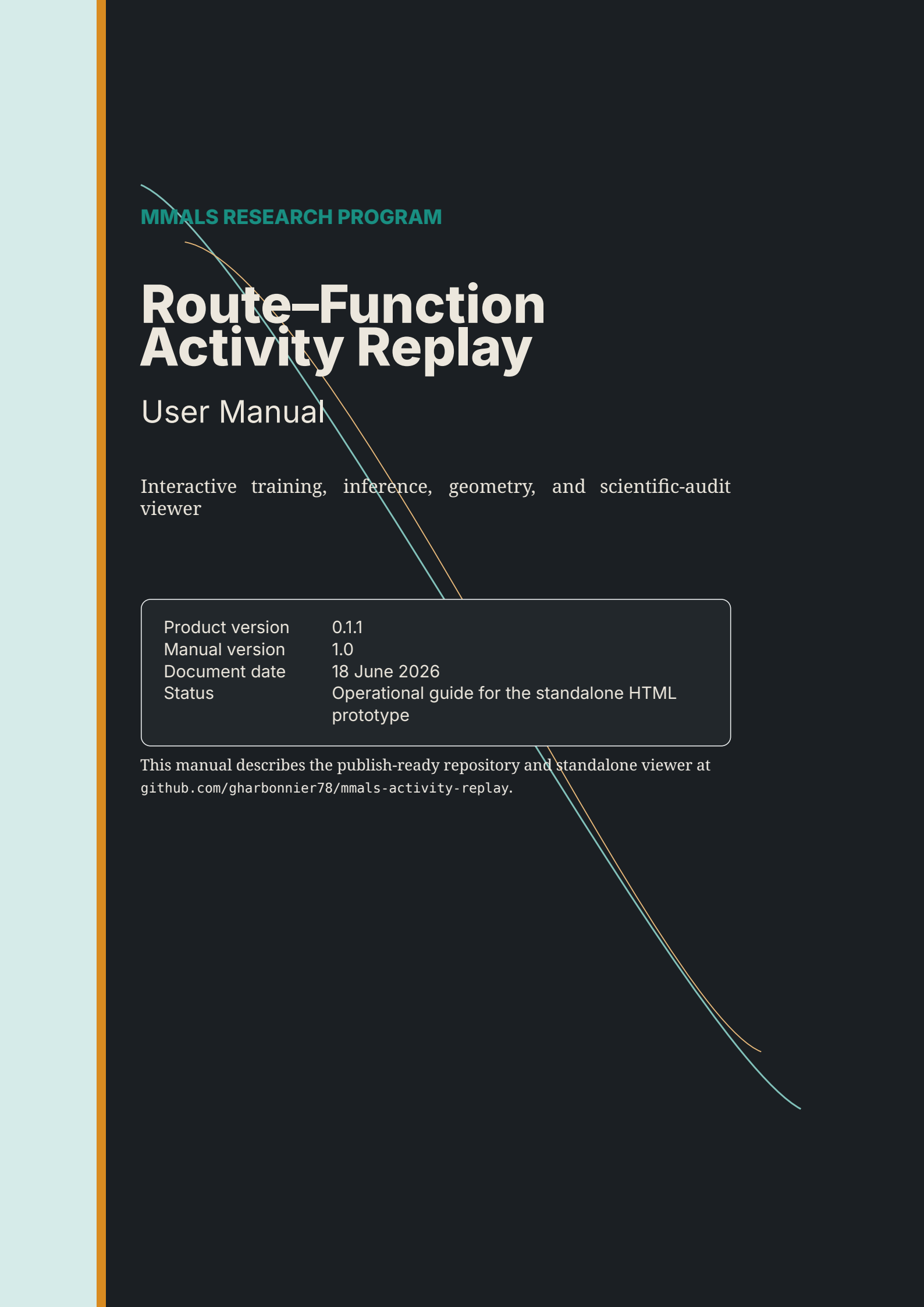




MMALS RESEARCH PROGRAM

Route-Function Activity Replay

User Manual

Interactive training, inference, geometry, and scientific-audit viewer

Product version	0.1.1
Manual version	1.0
Document date	18 June 2026
Status	Operational guide for the standalone HTML prototype

This manual describes the publish-ready repository and standalone viewer at github.com/gharbonnier78/mmals-activity-replay.

At a glance

MMALS Route–Function Activity Replay is a local, browser-based replay and audit tool for MMALS experiment traces. It synchronizes an ecosystem graph, a route-simplex view, metric cards, time-series traces, event details, and an A-to-B comparison. The embedded demonstration uses MMALS v0.9 CSV exports. Compatible CSV files can be loaded directly from the user’s computer; the viewer does not upload them.

Scope boundary

The viewer is an observational instrument. It visualizes logged measurements and clearly marked proxies. It does not retrain a model, recompute routes, establish causality, or convert routing entropy into physical carbon emissions unless explicit energy or carbon columns are present.

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1 Purpose and Scope

1.1 What the viewer is for

Use the viewer to replay two complementary forms of MMALS activity:

- **Training activity:** task and total losses, routing entropy, dominance, mutualistic gain, route-memory loss, distillation loss, and latent-memory loss.
- **Inference and audit activity:** retained-task accuracy, route allocation, route drift, latent functional drift, output drift, and checkpoint-to-checkpoint changes.

The synchronized interface helps answer practical questions such as:

- When did a route become specialized?
- Did the selected route remain stable while the mediated function drifted?
- Did accuracy remain stable even though internal geometry changed?
- Which hosts gained or lost routing weight between two checkpoints?
- Is a displayed “carbon” value physical, or only the historical entropy proxy?
- Is a regime-change value a calibrated detector output, or an uncalibrated visual diagnostic?

1.2 Disciplined biological mapping

The interface preserves the MMALS mapping used throughout the research program:

Biological term	MMALS element	Meaning in the viewer
Host organism / tree	Host neural module	A specialized representation-producing component, shown as a distinct node.
Fungal system / mycelial medium	Router and host-specific transformation system	The active mediator that combines, transforms, and allocates host signals.
Mycorrhiza	Host–medium exchange relation	The edges between hosts and the fungal medium; not a separate host and not the medium itself.
Carbon cost	Routing or physical resource cost	Routing entropy by default; physical energy or CO ₂ e only when explicitly logged.
Phosphorus / nutrient gain	Mutualistic contribution signal	Logged mean/minimum gain where available; not automatically a causal per-host estimate.

1.3 Why route geometry is a simplex, not a torus

For M hosts, route weights satisfy

$$r_i \geq 0, \quad \sum_{i=1}^M r_i = 1.$$

They therefore live on the probability simplex Δ^{M-1} . The viewer projects the five-host simplex to a two-dimensional barycentric display. A torus would be structurally justified only if the source variables contained two independent periodic coordinates. It is not imposed merely because toroidal geometry appears in other biological or artificial systems.

Interpretation rule

Treat the route-simplex point as a compact visualization of *allocation*. Do not treat its apparent two-dimensional distance as an exact metric on the full four-dimensional five-host simplex unless a specific projection and distance have been validated for that purpose.

2 Package Contents and Requirements

2.1 Files in the prototype package

File	Purpose
index.html	Standalone interactive viewer and GitHub Pages entry point.
README.md	Brief package description, fidelity boundaries, and recognized CSV families.
losses_df.csv	Training checkpoints and losses used by the embedded v0.9 demonstration.
history_df.csv	Post-task accuracy, route entropy, and dominance history.
routes_df.csv	Per-host route weights at logged checkpoints.
drift_df.csv	Route, latent, and output drift diagnostics.
task_forgetting_df.csv	Best/final task accuracy and per-task forgetting.
model_diag.csv	Model-level summary diagnostics.
scorecard.csv	Final accuracy, forgetting, variability, and parameter counts.
mmals_event_trace_schema.csv	Recommended long-format schema for a future event-level activity movie.

2.2 System requirements

- A current desktop browser with JavaScript enabled: Chromium/Chrome, Microsoft Edge, Firefox, or Safari.
- No web server is required for normal use.
- No network connection is required after the file is available locally.
- Sufficient browser memory for the selected CSV set. Very large event traces should be pre-filtered or sampled.

Local-data behavior

CSV files selected with **Load CSV files** are read by browser-side JavaScript. The prototype contains no upload function and sends no data to a remote service.

3 Quick Start

First replay in five steps

1. Extract the GitHub repository package to a local folder.
2. Open `index.html` in a modern browser.
3. Leave **Mode** set to **Training activity** and select a model/policy and seed.

4. Press **Play**; adjust **Playback speed** as needed.
5. Move the time slider to any checkpoint and inspect the ecosystem graph, route geometry, metric cards, traces, and current-event table together.

3.1 Fast inference/audit workflow

1. Change **Mode** to **Inference / audit**.
2. Select the observed model/policy and seed.
3. Enable the trace buttons for **accuracy**, **route drift**, **latent drift**, and **output drift**.
4. Move to a checkpoint of interest and press **Pin current as A**.
5. Move to a later checkpoint B and read the A-to-B delta panel.

Do not infer causality from one replay

An A-to-B change shows association within the selected trace. A causal objective, route-swap, or ablation claim requires a controlled intervention and a separately logged run or event identifier.

4 Interface Overview

4.1 Page map

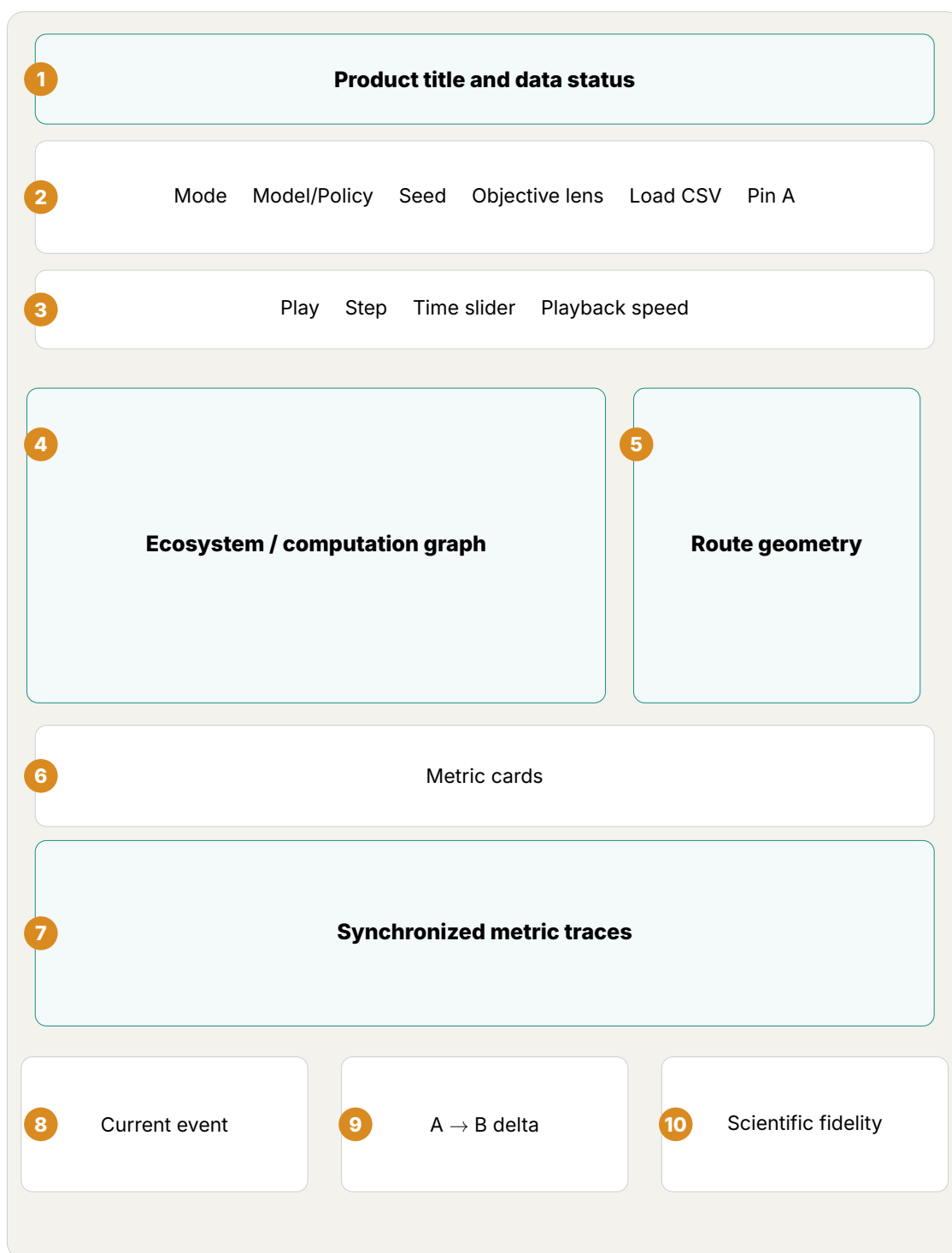


Figure 1: Logical page map. All panels are synchronized to the same selected replay step.

No.	Area	Function
1	Header/status	Identifies the viewer and confirms whether the embedded demo or local files are active.
2	Selection controls	Chooses mode, model/policy, seed, objective lens, data source, and A snapshot.
3	Replay controls	Starts, pauses, steps, scrubs, and changes playback speed.
4	Ecosystem graph	Displays component activity and forward/backward or memory relations using the MMALS biological mapping.
5	Route geometry	Displays the current host-allocation point and its fading trail in a five-vertex simplex projection.
6	Metric cards	Displays raw current values and a short statement about their source or proxy status.
7	Synced traces	Displays selected normalized time-series lines aligned to the same playhead.
8	Current event	Lists the active checkpoint's model, task, phase, route, losses, drifts, and related fields.
9	A-to-B delta	Compares a pinned snapshot A with the current snapshot B.
10	Scientific fidelity	States whether carbon, gain, regime, route time, and objective information are measured, approximated, or unavailable.

5 Controls

5.1 Mode

Training activity builds the replay primarily from a loss/checkpoint table containing fields such as `task_id`, `epoch`, and `task_loss`. It emphasizes training losses, route specialization, entropy, mutualistic gain, and memory terms.

Inference / audit combines evaluation history, route checkpoints, and drift diagnostics. It emphasizes retained-task accuracy and internal drift after sequential learning.

5.2 Observed model / policy

The viewer constructs this list from the `model` or `method` column across recognized input tables. Changing the model rebuilds the replay and refreshes the available seeds.

5.3 Seed

The seed selector filters the selected model to one experimental realization. If the seed column is absent, the viewer treats the data as seed 0.

5.4 Objective lens

The objective lens changes visual emphasis and a composite display score. Available lenses are:

- **Balanced ecosystem**
- **Performance**
- **Stability**

- **Mutualistic ecology**
- **Efficiency / carbon**

Lens is not intervention

Changing the lens does not recompute the route, model output, or training objective. It only changes how the already observed state is visually emphasized. Real objective-change analysis requires separate logged runs identified by `objective_id` or an explicit goal vector.

5.5 Load CSV files

Select one or several compatible CSV files. The viewer classifies files by their columns rather than by exact filename. The newly selected files replace the embedded dataset for the current browser session.

5.6 Pin current as A

Press this button to store the current state as comparison snapshot A. Continue the replay or move the slider to another state B. The A-to-B panel then displays deltas in available metrics and route weights.

5.7 Play, Step, time slider, and speed

Control	Behavior
Play / Pause	Advances through the ordered states or pauses at the current state.
Step	Advances by one logged state. Useful for checkpoint-level inspection.
Time slider	Jumps directly to a replay index. All views update together.
Playback speed	Changes the delay between states from approximately $0.25\times$ to $8\times$. It does not alter the underlying timestamps.

6 Main Visual Views

6.1 Ecosystem / computation graph

The graph provides a stable component topology while metric values vary over time.

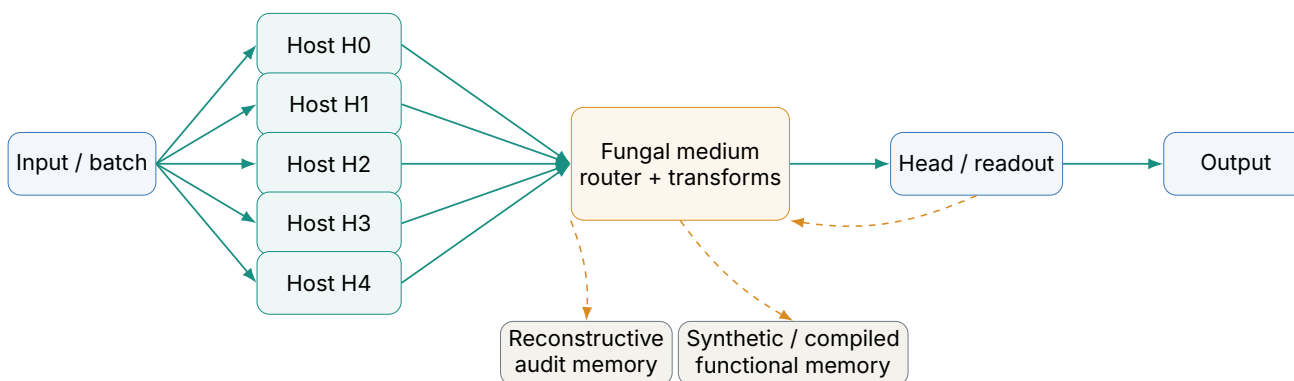


Figure 2: Conceptual ecosystem graph used to interpret the replay.

Typical visual encodings include:

- host node intensity or size: current route allocation and available gain signal;
- edge thickness: relative exchange strength;

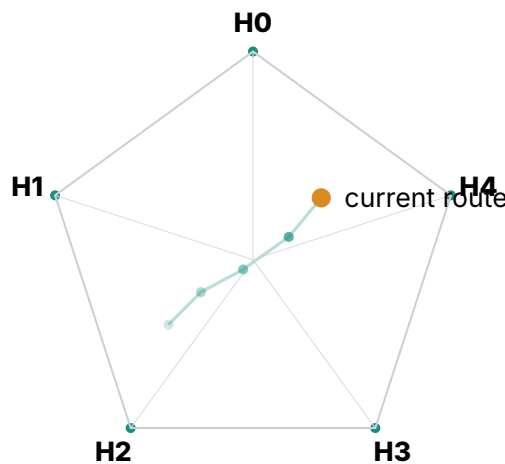
- direction/style: forward activity, backward learning, or memory read/write;
- fungal-medium activity: mediated route/function state;
- objective node: display-lens score, not a causal change to the observed run.

6.2 Route geometry

For five hosts, the current route is a vector

$$r = (r_0, r_1, r_2, r_3, r_4), \quad r_i \geq 0, \quad \sum_i r_i = 1.$$

The viewer maps each host to one vertex of a regular pentagon and places the route point at the weighted average of the vertices.



The fading trail shows successive logged states, not a continuous physical path unless the trace is event-level.

Figure 3: Five-host route projection with a fading replay trail.

A point close to a vertex indicates strong routing dominance by that host. A point near the center indicates a more diffuse allocation. The trail helps reveal route movement, but checkpoint spacing may be uneven.

6.3 Metric cards

Cards show raw current values, usually including:

- task loss or accuracy;
- route dominance;
- carbon / entropy;
- phosphorus / mutualistic gain;
- route, latent, or output drift;
- regime-change probability or proxy.

The explanatory text below each value is as important as the value itself. It identifies whether the metric was logged directly, derived from an available field, or substituted by a proxy.

6.4 Synchronized traces

Trace buttons turn individual metrics on or off. Each active line is normalized within the selected model/seed replay so that metrics with different units can share one chart. Raw values remain in cards and tooltips.

Normalization boundary

Do not compare the vertical amplitude of two normalized lines as if their original units were identical. Use normalized traces to compare timing and direction; use cards, event tables, or exported CSV values for quantitative comparisons.

6.5 Current event

The event table exposes the selected replay state's underlying values. It is the preferred place to confirm task identifiers, after-task index, epoch, phase, route status, and raw drift or loss fields.

6.6 A-to-B delta

After snapshot A is pinned, the panel compares A with the current state B. Available deltas include loss, accuracy, carbon/entropy, gain, and host route weights. A positive route delta means a host received more weight at B; a negative value means less.

6.7 Scientific fidelity panel

This panel is designed to prevent attractive graphics from becoming stronger scientific claims than the data support. It reports the status of:

- carbon/energy measurement;
- phosphorus or mutualistic gain;
- regime-change probability;
- route timing and interpolation;
- objective changes.

7 Training Replay Workflow

Recommended training inspection

1. Select **Training activity**, model/policy, and seed.
2. Start with **task loss**, **carbon / entropy**, **phosphorus gain**, and **regime change** traces active.
3. Step through the first task and note when route dominance increases and entropy decreases.
4. At each task boundary, inspect whether the route point moves abruptly or gradually.
5. Enable memory-related traces or inspect the event table for route-memory, distillation, and latent losses.
6. Pin A before an expected transition, move to B after the transition, and compare route allocation and losses.

7.1 How to read common training metrics

Metric	Typical field	Interpretation
Task loss	task_loss	Supervised loss for the current task or batch checkpoint. Lower is usually better within a comparable run.
Total loss	total_loss	Task loss plus active MMALS regularization/memory terms. It should not be compared directly across methods with different objectives without decomposition.
Entropy / carbon proxy	entropy or route_entropy	Diffuseness of route allocation. Lower entropy means more concentrated routing, not automatically lower physical energy.
Dominance	dominance or route_dominance	Maximum route weight. High dominance indicates concentrated allocation, but not necessarily useful specialization.
Mutualistic gain	gain_mean, gain_min	Aggregate contribution signal. A positive mean may coexist with a negative minimum. Per-host causality requires host-level ablation or gain fields.
Route-memory loss	route_memory_loss	Penalty for changing preserved route patterns. Low route drift does not guarantee low functional drift.
Distillation loss	distill_loss	Difference between current and stored output behavior.
Latent loss	latent_loss	Difference between current and stored mediated latent function.
Warm-up coefficient	alpha	Activation/ramp factor for delayed mutualistic or memory terms in compatible notebooks.

7.2 A useful diagnostic pattern

A stable route with rising latent drift is important:

$$D_r \approx 0 \quad \text{while} \quad D_z > 0.$$

It means the route identity remained similar while the computation through the route changed. This is a central route–function distinction in MMALS and is exactly the kind of pattern the viewer is intended to expose.

8 Inference and Audit Replay Workflow

Recommended inference/audit inspection

1. Select **Inference / audit**.
2. Activate **accuracy**, **dominance**, **latent drift**, and **regime change** traces.
3. Move through each after_task checkpoint.
4. Check whether old-task accuracy changes after later tasks are learned.
5. Compare route drift D_r , latent drift D_z , and output drift D_y rather than reading only

final accuracy.

6. Pin a pre-change state A and compare it with a later state B.

8.1 Drift hierarchy

The prototype supports three complementary drift families:

$$\begin{aligned} D_r &= \mathbb{E} \|r_{\text{current}} - r_{\text{reference}}\|_2^2, \\ D_z &= \mathbb{E} \|z_{f,\text{current}} - z_{f,\text{reference}}\|_2^2, \\ D_y &= \mathbb{E} D_{\text{KL}}(p_{\text{reference}} \| p_{\text{current}}). \end{aligned}$$

Drift	Question	Caution
D_r	Did host allocation change?	Route stability can be cosmetic if host functions or transforms drift.
D_z	Did the mediated latent computation change?	Latent distance depends on representation scale and alignment.
D_y	Did observable predictive behavior change?	Output stability can hide internal reorganization and may be task/head dependent.

8.2 Accuracy and forgetting

The history table may contain post-task accuracy $A_\tau^{(s)}$: performance on task τ after learning through task s . Per-task forgetting is commonly summarized as

$$F_\tau = \max_{s \geq \tau} A_\tau^{(s)} - A_\tau^{(K)}.$$

The viewer can display the checkpoints used to understand this value, but it does not replace the experiment’s statistical aggregation across seeds.

9 Metric Dictionary and Scientific Meaning

9.1 Route entropy and dominance

For route r over M hosts,

$$H(r) = - \sum_{i=1}^M r_i \log(r_i + \varepsilon), \quad \text{dominance}(r) = \max_i r_i.$$

Low entropy and high dominance indicate concentrated routing. They do not, by themselves, prove host specialization. Stronger evidence requires contribution gain, ablation impact, representation separation, route–function stability, and cross-seed reproducibility.

9.2 Carbon

The viewer uses the following precedence:

1. an explicit recognized field such as `carbon_cost`, `co2e_g`, `energy_j`, or `energy_kwh`;
2. otherwise, routing entropy as the historical MMALS “carbon tax” proxy.

Units must remain visible

An energy value in joules, an energy value in kilowatt-hours, and a CO₂e value in grams are different quantities. The event-level schema keeps them in separate columns. Do not merge them into a single unlabeled “carbon” score for scientific reporting.

9.3 Phosphorus / mutualistic gain

The demonstration recognizes `gain_mean` and `gain_min`. Conceptually, a host contribution can be estimated by removing host i and comparing loss:

$$G_i = \mathcal{L}_{\text{without } i} - \mathcal{L}_{\text{full}}.$$

A positive G_i means removal made the loss worse, so the host contributed under that intervention. The current v0.9 dump does not contain a causal G_i for every host at every event. Consequently, host glow is driven mainly by route allocation and aggregate gain, not by a complete per-host causal estimate.

9.4 Regime-change display

If `regime_prob`, `regime_change_prob`, or `context_change_prob` is present, the viewer displays that value as a logged detector output.

Otherwise, it constructs an uncalibrated visual proxy from changes in route, loss, drift, and known task boundaries. The proxy is useful for locating possible transitions, not for claiming a calibrated probability.

9.5 Composite objective score

The ecosystem graph may display a lens-dependent score combining performance, stability, ecology, and efficiency signals. This score is a visualization aid. It is not a contractual evaluation metric unless the research protocol explicitly freezes its formula and thresholds.

10 Loading Your Own CSV Data**10.1 General rules**

- Select all related CSV files in one file-picker action.
- Column names, not filenames, determine how tables are classified.
- The model identifier may be named `model` or `method`.
- Empty cells and the text `NaN` are treated as missing values.
- Numeric text is converted to numbers when possible.
- The newly loaded set replaces the embedded set until the page is reloaded.

10.2 Recognized notebook-style families

Family	Minimum identifying columns	Useful optional columns
Training losses	<code>task_id</code> , <code>epoch</code> , <code>task_loss</code>	<code>total_loss</code> , <code>entropy</code> , <code>dominance</code> , <code>gain_mean</code> , <code>memory losses</code> , <code>alpha</code>
Routes	one or more <code>host_0</code> , <code>host_1</code> , ...columns	<code>after_task</code> , <code>task_id</code> , <code>task_name</code>

Family	Minimum identifying columns	Useful optional columns
Evaluation history	after_task, task_id, acc	route_entropy, route_dominance
Drift diagnostics	route_drift, latent_drift, output_drift	task and model identifiers
Newer drift names	route_drift_Dr, latent_drift_Dz, output_drift_Dy	transformation/Jacobian drift fields
Explicit regime	any supported regime-probability field	true/predicted regime labels
Physical cost	any explicit energy or CO ₂ e field	latency, active parameters, hardware metadata

10.3 Example: minimum training CSV

Listing 1: Minimal training replay table

```
seed,model,task_id,task_name,epoch,task_loss,total_loss,entropy,dominance
0,example_policy,0,0 vs 1,0,0.52,0.58,1.31,0.34
0,example_policy,0,0 vs 1,1,0.28,0.33,0.91,0.55
0,example_policy,1,2 vs 3,0,0.61,0.75,1.22,0.41
```

10.4 Example: route CSV

Listing 2: Five-host route checkpoints

```
seed,model,after_task,task_id,task_name,host_0,host_1,host_2,host_3,host_4
0,example_policy,0,0,0 vs 1,0.62,0.08,0.12,0.10,0.08
0,example_policy,1,0,0 vs 1,0.58,0.07,0.16,0.11,0.08
```

10.5 Data-preparation checks

Before loading, verify:

- route weights are numeric and preferably sum to approximately 1;
- task identifiers use consistent numeric types;
- model and seed labels are consistent across files;
- after_task and task_id mean the same thing in all tables;
- physical cost columns include units in their names;
- regime probabilities are bounded to $[0, 1]$ and are actually calibrated if described as probabilities.

11 A-to-B Comparison

11.1 Basic use

1. Navigate to the baseline checkpoint.
2. Press **Pin current as A**.
3. Move to another replay state B.

4. Read the delta summary and per-host route bars.
5. Record the model, seed, task, and replay indices before using the comparison in a report.

11.2 Good use cases

- before versus after a task boundary;
- early versus late epoch of one task;
- an old task immediately after learning versus after later tasks;
- a low-entropy checkpoint versus a high-entropy checkpoint;
- a suspected regime transition versus a stable neighborhood.

11.3 Invalid or misleading use cases

- comparing different objective lenses and calling the visual difference a causal intervention;
- comparing states from different models without separately selecting and exporting them;
- treating normalized chart height as a raw effect size;
- calling a route-weight increase “host benefit” without contribution or ablation evidence;
- treating the uncalibrated regime proxy as a probability estimate.

12 Scientific Fidelity Boundaries

Boundary	Operational meaning
Route time	v0.9 contains route checkpoints rather than every optimizer step. Training routes may be transparently interpolated between a uniform start and a logged checkpoint.
Carbon	Routing entropy is used unless explicit physical energy or CO ₂ e columns exist.
Phosphorus gain	Aggregate mean/minimum gain may be logged; per-host causal gain is not reconstructed from aggregate data.
Regime probability	A calibrated value is shown only when a recognized probability column exists. Otherwise the viewer labels its value as a change proxy.
Objective changes	Lens selection changes visual emphasis only. Counterfactual objective effects require separate experimental runs.
Simplex projection	The two-dimensional pentagonal display is a projection of a higher-dimensional route simplex, not a proof of intrinsic two-dimensional geometry.
Graph topology	The component graph is a stable explanatory layout. Node position is not automatically a learned embedding.
Color and glow	Visual intensity is an aid to attention, not an independent measurement.

Recommended statement for papers or demonstrations

“The viewer is used as a synchronized exploratory and audit interface over logged MMALS traces. Quantitative claims are computed from exported data and protocol-defined met-

rics; visual encodings are not treated as independent evidence.”

13 Troubleshooting

Symptom	Resolution
Blank page or controls do not respond	Confirm JavaScript is enabled. Reopen the file in a current browser. Check the browser developer console for a syntax or memory error.
The browser blocks local behavior	Move the complete package to a normal local folder. As a fallback, start a local server in that folder with <code>python -m http.server 8000</code> and open <code>http://localhost:8000/index.html</code> .
Model selector is empty after loading	At least one recognized table must contain a model or method value and identifying metric columns. Confirm headers have no hidden characters.
Seed selector shows only 0	The seed column is absent, empty, or every row uses seed 0.
No route geometry appears	Load a route table with numeric <code>host_0</code> , <code>host_1</code> , ...columns matching the selected model and seed.
Route weights look incorrect	Check that weights are nonnegative and correspond to the same host order in every file. The viewer normalizes available route weights for display.
Carbon card shows entropy	No supported explicit energy/carbon field was found. Add <code>energy_j</code> , <code>energy_kwh</code> , <code>co2e_g</code> , or <code>carbon_cost</code> .
Gain is missing	The selected state has no <code>gain_mean</code> / <code>gain_min</code> data. This is common in inference-only exports.
Regime value says “change proxy”	No supported logged probability field is present. Add a real detector output under a recognized column name.
Replay jumps between sparse checkpoints	The source table is checkpoint-level. Export event-level or more frequent state rows for smoother replay.
Page becomes slow with large files	Pre-filter by model/seed/task, down-sample high-frequency rows, or split event traces. The current prototype loads all selected rows into browser memory.
A-to-B panel seems inconsistent	Re-pin A after changing mode, model, seed, or data set. Those changes rebuild the replay state.
Normalized traces look flat	A metric may have little variation or only one non-missing value. Inspect raw cards/event values.

14 Recommended Event-Level Trace

Notebook-style CSV dumps are sufficient for checkpoint replay. A true architecture-agnostic activity movie is better represented by a long-format event trace: one row per component or exchange event, globally ordered by step.

14.1 Core event groups

Group	Recommended fields
Ordering and time	<code>step</code> , <code>wall_time_s</code> , <code>mode</code> , <code>phase</code>
Experiment identity	<code>model</code> , <code>backbone</code> , <code>seed</code> , <code>objective_id</code> , <code>task_id</code>

Group	Recommended fields
Context/regime	regime_true, regime_pred, regime_prob
Interaction	src_component, dst_component, direction, signal_type, exchange_value
Component state	component_activity, host_id, route_weight
Specialization	host_gain, host_ablation_delta, route_entropy, route_dominance
Performance and memory	task_loss, memory_loss, accuracy
Geometry and drift	route_drift, latent_drift, output_drift, transform_drift
Calibration	calibration_width
Operational cost	latency_ms, active_params, energy_j, energy_kwh, co2e_g
Legacy proxy	carbon_proxy_entropy
Provenance	notes plus run/commit/config identifiers in future extensions

14.2 Direction vocabulary

Recommended controlled values include:

- forward: activations or mediated signals moving toward prediction;
- backward: gradient or credit-assignment signals;
- memory_read: retrieved reconstructive or compiled memory;
- memory_write: creation/update of memory artifacts;
- audit: measurement-only events that do not affect model state.

14.3 Architecture-independent components

Component identifiers can cover MLPs, CNNs, transformer blocks, routers, heads, adapters, memories, and external tools. Examples:

Listing 3: Example component identifiers

```
input.batch
encoder.cnn.block_2
encoder.transformer.layer_5.attention
host.0
host.3
fungal.router
fungal.transform.3
memory.reconstructive
memory.compiled
head.task_2
output.logits
```

15 Suggested Research Workflows

15.1 Route–function gap audit

1. Select inference/audit mode.
2. Trace D_r , D_z , D_y , and accuracy.
3. Locate intervals with low D_r but high D_z .

4. Compare the same tasks and seeds across route-only and functional-memory methods outside the viewer using the exported data.

15.2 Host-specialization audit

Use the viewer to locate candidate specialization events, then confirm them with:

- route entropy and dominance;
- host contribution gain;
- host ablation delta;
- representation separation;
- route–function stability;
- candidate-permutation controls;
- cross-seed reproducibility.

15.3 Regime-change detector audit

1. Export the detector’s actual `regime_prob`.
2. Display it next to route, latent, output, and performance traces.
3. Mark ground-truth regime boundaries separately.
4. Evaluate calibration, detection delay, false alarms, and downstream action policy outside the visual replay.

15.4 Objective-change experiment

To study a real objective intervention rather than a display lens:

1. freeze the data split, host bank, initialization policy, and evaluation protocol;
2. run each objective under a unique `objective_id`;
3. log goal-vector values and all affected costs;
4. compare routes, functions, host contribution, accuracy, forgetting, and operational cost across matched seeds;
5. use the viewer to replay each run, then use protocol-defined statistics for the claim.

16 Frequently Asked Questions

Does the viewer prove that MMALS works?

No. It helps inspect and communicate experiment traces. Evidence comes from controlled protocols, correct baselines, repeated seeds, statistical analysis, and reproducible artifacts.

Why is there no torus view?

The route is a simplex variable. A torus should be added only for genuinely periodic coordinates or validated toroidal latent structure.

Can it replay CNNs, transformers, routers, and heads?

Yes, conceptually, when they are represented as component and exchange events. The current v0.1 notebook-style loader is optimized for MMALS checkpoint tables; the event-level schema is the path to generalization.

Does the efficiency lens change the selected route?

No. It changes visual emphasis only.

Is routing entropy an estimate of CO₂ emissions?

No. It is a model-level proxy for diffuse routing. Physical energy and CO₂e require explicit measurements and system-boundary documentation.

Does a glowing host mean it is causally useful?

Not necessarily. In the current demonstration, glow is driven by route allocation and aggregate gain availability. Causal usefulness needs host-level intervention evidence.

Are local CSV files uploaded?

No. They are parsed locally in the browser by the standalone page.

Can I use screenshots in a paper?

Yes, as explanatory figures, provided the caption states that the graphic is a visualization of logged traces and identifies measured versus proxy values. Numerical claims should cite the underlying tables and protocol.

A One-Page Operator Checklist

Before opening

- Extract the complete package.
- Keep related CSV files together.
- Confirm browser JavaScript is enabled.

Before interpreting

- Confirm mode, model/policy, and seed.
- Read the scientific-fidelity panel.
- Check whether carbon and regime values are measured or proxies.
- Confirm whether the replay is checkpoint-level or event-level.

Before comparing A and B

- Pin A after the correct replay rebuild.
- Record A and B step/task identifiers.
- Inspect raw values, not only normalized traces.
- Do not call a visual delta causal without an intervention.

Before publishing

- Export and archive source CSVs, configuration, seed, and commit identifier.
- State proxy definitions and units.
- Validate claims across matched controls and seeds.
- Use the viewer as an audit/explanation layer, not as the sole evidence source.

B Color and Symbol Legend

Color/symbol	Typical use	Meaning
 Teal	Hosts, routes, forward exchange	Active allocation or mediated signal.
 Amber	Objective, warnings, current route	Attention, selected state, or cost-oriented signal.
 Blue	Outputs, information, audit aids	Readout, information, or explanatory status.
	Confirmed / available	Logged or operational state is available.
	Warning / proxy	Interpretation requires a boundary or caveat.
	Information	Explanatory or non-causal display element.

C Version Notes

Version	Date	Notes
Viewer 0.1.1	June 2026	Standalone HTML prototype; embedded v0.9 demonstration; training/inference modes; ecosystem graph; route-simplex geometry; synced traces; A-to-B comparison; local CSV loading; fidelity panel.
Manual 1.0	18 June 2026	Initial operational manual covering controls, data schemas, metrics, scientific boundaries, troubleshooting, and future event-level trace.

End of manual

MMALS Route–Function Activity Replay v0.1.1